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Recycled and upcycled materials in contemporary architecture and civil engineering: Their applications, benefits, and challenges

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ABSTRACT

This paper explores the increasingly popular practice of integrating recycled and upcycled materials in modern architecture and civil engineering and highlights their central role in promoting sustainability, energy efficiency, and environmental responsibility. Upcycling transforms discarded materials, such as reclaimed metal, wood, masonry, glass, concrete, and plastic, into valuable resources and offers a promising course for achieving sustainability. This research evaluates the performance, durability, and aesthetic potential of discarded materials through case studies of completed projects to illustrate their successful applications. It also addresses the challenges and opportunities associated with the widespread use of recycled and upcycled materials by focusing on technological advancements, lifecycle assessments, and economic feasibility. The findings underscore the significance of continued research and innovation in this field and the need for supportive government regulations and better public awareness. In summary, the study emphasizes the compelling need to integrate recycled and upcycled materials in construction to promote sustainable construction practices and develop a more resilient, resource-efficient built environment.

1. Introduction

In 2015, the United Nations launched the 2030 Agenda for Sustainable Development and introduced the Sustainable Development Goals (SDGs) to tackle global challenges and promote a more sustainable future by 2030 (Xu et al., 2024). Architects and civil engineers respond to the growing universal concerns about sustainability, environmental impact, and resource scarcity by exploring creative solutions (Hodge et al., 2010). One key strategy in creative solutions is integrating recycled and upcycled materials into construction. This shift in strategy addresses environmental issues and paves the way for new opportunities for efficiency and creativity in design and construction. As a result, the construction industry gradually shifts to using recycled and upcycled materials to address resource scarcity and environmental concerns (Mhatre et al., 2021).

Among the materials created when transforming waste into new materials are recycled concrete, metals, glasses, and plastics. Repurposed materials retain their inherent qualities when used for new purposes, often with higher value and consuming less energy, such as when constructing structures using old shipping containers and laying down wood pallets for flooring (Joensuu et al., 2020). Table 1 presents the primary differences between recycling and upcycling and their processes, outcomes, and environmental implications (Mazur, 2021).

Production of recycled and upcycled materials often requires less energy than manufacturing new materials, thus reducing carbon emissions. Using recycled and upcycled materials in resource- and environmentally friendly construction techniques aligns with sustainable design principles (Hole and Hole, 2019; Zhao et al., 2022). From an economic perspective, constructions that utilize recycled and upcycled materials are more cost-effective and offer market advantages (Bridgens et al., 2018). They also provide a competitive edge since constructions with sustainable materials are more appealing to eco-aware clients (Aguome et al., 2024).

Repurposing materials encourages creativity in solving problems and design, occasionally producing odd and aesthetically stunning structure designs. The financial benefits of using recycled and reused materials are the appealing options for modern construction projects (Nian et al., 2022). Some areas of civil engineering and architecture use recycled or upcycled materials (Kuoribo et al., 2024; Nicolas and Filz, 2024; Santos et al., 2024), including using recycled steel and concrete as structural elements in buildings, bridges, and roads to preserve their original strength and durability. The excellent thermal properties of recycled plastic and glass make them suitable for fabricating roofing materials and insulation. Reclaimed wood and metal are examples of upcycled

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Table 1
Differences between upcycling and recycling.
(Kosloski-Oh et al., 2021; Yi et al., 2018).

Criteria	Upcycling	Recycling
Definition	Convert waste resources into more valuable or superior goods.	Convert waste resources into new, comparable, or less valuable goods.
Process	Minimum processing and creative material reuse.	Reprocess the materials after disassembling them into their most basic forms.
Product Value	Superior to or similar quality to the original material.	Often have less or the same value as the source material.
Ecological Impact	Minimize waste while conserving resources and energy.	Minimize waste but require reprocessing energy.
Illustration	Transform used furniture into fresh, original designs.	Use melted plastic bottles to create new plastic products.
Resource Efficiency	Leverage current goods with minor modifications to make them more resource efficient.	Resource-intensive because it uses energy to break down the materials.
Degree of Innovation	High innovation since it depends on creative, imaginative, and inventive design.	Moderate since it emphasizes recovering material above design.
Raw Materials Usage	Use pre-existing products and require no or minimal raw material procurement.	Require constituent components (additives) or additional resources to complete the recycling processes.

materials utilized in interior design, such as flooring, furniture, and decorative elements, since they add character with minimal environmental impact (Galluccio et al., 2024; Guerra and Leite, 2021a). Recycled materials are increasingly used in urban infrastructure projects, such as recycled plastic in park benches and playground equipment and broken concrete in the layers of road foundations (Bakker et al., 2010). Despite their benefits, there are hindrances to using recycled and repurposed materials (Fayed and Mansour, 2023). It is crucial to provide information and evidence of the benefits and reliability of recycled and upcycled alternatives to dispel market skepticism and stakeholders' reluctance since they are used to utilizing traditional materials (Guselnikova et al., 2023). This systematic study will analyze the current research and practice of integrating recycled and upcycled materials in modern architecture and civil engineering.

The review objective is to gather and synthesize existing literature in an organized manner to determine the significant trends and gaps, assess the social, economic, and environmental benefits of recycled and upcycled materials, investigate the hindrances to their adoption, evaluate skilled case studies, and provide constructive recommendations for stakeholders. The review also proposes avenues for further research to gain better insights and implement them in the field. In summary, this review hopes that the improved insights could facilitate the transition towards more environmentally friendly construction techniques. Recycling and upcycling minimize the volume of materials disposed of in landfills. Recycling is a practical industrial process focusing on eco-efficiency to reduce the adverse effects of production. Upcycling is a highly creative process that optimizes the positive impact of reuse and emphasizes eco-effectiveness.

2. Incorporating recycled and upcycled materials in architectural designs

Architects are moving towards incorporating repurposed and recycled materials in their projects to promote sustainability, innovation, and environmental responsibility (Papamichael et al., 2023). Besides doing so as a moral choice, they use recycled and upcycled materials imaginatively in architectural designs. As the construction industry advances, architects and civil engineers use recycled and upcycled materials to design innovative, sustainable, and aesthetically pleasing buildings. Possible goals for architects who prioritize reusing and recycling materials are to redefine modern architecture and achieve a more sustainable future (Huang et al., 2024). Using recycled and upcycled materials in architectural projects is also a significant step toward a more resource-efficient and sustainable built environment. Architects are pushing creative and aesthetically pleasing solutions by using materials like glass, metals, concrete, and old furniture creatively to reduce their environmental impacts. It is imperative to adopt these strategies in addressing environmental concerns and producing innovative and practical architectural designs as the construction industry grows. With more in-depth research and collaboration, architects will be able to blend creativity with sustainability in the future, thus leading to a more responsible built environment that ultimately supports a healthier global community (Bao, 2023).

2.1. Historical overview of using recycled and upcycled materials for sustainable construction

Architects and civil engineers have increasingly used recycled and upcycled materials in recent decades as they recognized the significance of adopting environmentally friendly practices and resource conservation (Sieffert et al., 2014). The literature review provides a historical overview of the development of techniques for using recycled and upcycled materials, highlighting significant breakthroughs, pivotal moments, and current cutting-edge technologies (Yu et al., 2024). The methods for incorporating recycled and upcycled materials in civil engineering and design change with time, demonstrating a growing dedication to sustainability and efficient use of resources (Kapica et al., 2024). The industry has progressed from using obsolete recycling techniques to pioneering innovations. With higher awareness and better technology, it is increasingly essential to integrate recycled and upcycled materials in future construction and contribute to creating a more resilient and sustainable built environment (Doğan-Sağlamtimur et al., 2018; Parece et al., 2022).

2.1.1. Early practices in recycling and upcycling materials

The idea of recycling in construction is not entirely new; historical evidence shows that ancient civilizations had reused materials long before modern engineering due to resource unavailability and the high cost of purchasing virgin resources (Burns, 2024). For instance, the Romans regularly recycled stones from ancient buildings to construct new ones to conserve resources and reduce construction expenses (Barker and Marano, 2017). Similarly, ancient Egyptians reused mud bricks and stones from earlier pyramids and temples to build new ones (El-desokey, 2023). Even though these practices are motivated primarily by needs, they provide the foundation for sustainability in construction. In the Middle Ages, European builders continued reusing components from destroyed structures in new constructions (Pryor, 1995). There were instances where they deconstructed whole structures and reused the components to construct new structures. This construction method was widely adopted during war and economic distress when resources were scarce, and recycling was a viable solution. However, systematic recycling and upcycling in architecture did not gain significance until the late 20th century, when industrialization led to mass manufacturing and consumption that generated high volumes of waste (Muñoz Velasco et al., 2014).

2.1.2. The industrial revolution and waste generation

The Industrial Revolution brought about a significant transformation in construction methods. Throughout the 19th century, rapid industrialization and urbanization dramatically increased the production and consumption of materials (Cipolla, 2004; Ngowi et al., 2005). Environmental concerns about industrial waste did not emerge until the early 20th century. The construction industry ignored recycling and upcycling during the first half of the century since its primary focus was cost-effectiveness, speed, and efficiency. Buildings were often designed

to be disposable and favored replacement over repair, reflecting the throwaway mentality in the construction industry (Cheng and Ma, 2013). The increasing waste generation accompanying the Industrial Revolution had significant environmental, social, and economic consequences, particularly in the European Union (EU-27). The increased waste generation contributed to air and water pollution and declining biodiversity. In response, the EU introduced various initiatives to mitigate waste generation. Despite these efforts, the per capita household waste in EU-27 has been steadily increasing since 2006, as shown in Fig. 1. By 2018, the highest recorded per capita waste generation was 174.5 kg per person, up from 161.2 kg per person in 2006. Since this trend is expected to continue due to the growing use of packaging materials across industries, it is crucial to implement monitoring and preventive measures to address the continuing increase in packaging waste (Axelsson-Bakri et al., 2020).

2.1.3. Environmental movement and the rebirth of recycling

The ecological awakening of the 1960s and 1970s resulted in significant shifts in the attitude towards sustainability and waste (Jewdokimow et al., 2024). The publication of *Silent Spring* in 1962, a book by Rachel Carson that highlights the dangers of industrial pollution, raised public awareness of environmental issues (Paull, 2013). Growing concerns about resource depletion and the destruction of the environment gave birth to modern recycling systems during this period and rekindled the interest in using recycled materials among architects and civil engineers (Li, 2024). Material reuse became increasingly economically viable as energy prices increased and oil prices spiraled upwards during the 1970s oil crisis. Architects and engineers began pioneering environmentally friendly building designs by incorporating recycled steel, glass, and concrete (Zabalza Bribián et al., 2011).

2.1.4. The late 20th century: from recycling to upcycling

Upcycling emerged in the latter part of the 20th century as sustainability became all the rage. Upcycling seeks to reuse materials in ways that increase their value, unlike recycling, which often involves breaking them down into their raw base components. This approach aligns with the principles of circular economy that prioritize waste reduction and optimal use of pre-existing resources (Hawthorne, 2014). Famed architects like Shigeru Ban and Frank Gehry are well-known for their creative use of recycled materials. For example, by incorporating corrugated metal and chain-link fences into his designs, Gehry challenged traditional notions of materiality and beauty in architecture. Shigeru Ban's use of cardboard and recycled paper tubes as structural components in disaster relief shelters (Rattenbury, 2004) demonstrates that repurposed materials have the potential to tackle global concerns.

2.1.5. Twenty-first-century innovations and practices

Technological advancements and the growing emphasis on sustainable development have markedly increased the utilization of recycled and upcycled materials in the 21st century. This shift is the consequence of, among others, technological advancements, environmental constraints, and increased awareness of the need for sustainable growth (Rašković et al., 2020). Two examples of green buildings constructed using recycled materials are the Building Research Establishment Environmental Assessment Methods (BREEAM) and Leadership in Energy and Environmental Design (LEED). Civil engineering projects can use recycled materials in more comprehensive ways because of the availability of new materials like recycled concrete aggregate (RCA) and reclaimed asphalt pavement (RAP) (Sieffert et al., 2014). In addition, digital fabrication techniques such as 3D printing facilitate using recycled materials in complex and customized building components. These materials function similarly to conventional materials but with less environmental impacts from construction activities. Upcycled materials are becoming more popular in architecture as designers incorporate them into creative and visually appealing constructions (C. Zhang et al., 2022). Architects can use repurposed materials to create contemporary, sustainable architecture. For example, the construction of The Crystal in London used primarily recycled materials, and The ECOARK in Taiwan used 1.5 million PET bottles for its wall. These materials are cost-effective substitutes that impart the structures with lightweight and excellent insulation properties (Yen et al., 2022).

2.1.6. Challenges and current trends

Despite the increasing use of new and repurposed materials in civil engineering and architecture, there are still obstacles to overcome. The major hindrance is recycled materials' unpredictable performance and quality, which may restrict their use in some applications. There is a need for further standardized testing and certification procedures to ensure their dependability and safety (Kapliński and Bonenberg, 2020; Plevris et al., 2022). In the future, it is crucial to have better industry cooperation and legislative frameworks that support recycled and upcycled materials and technological advancements in material

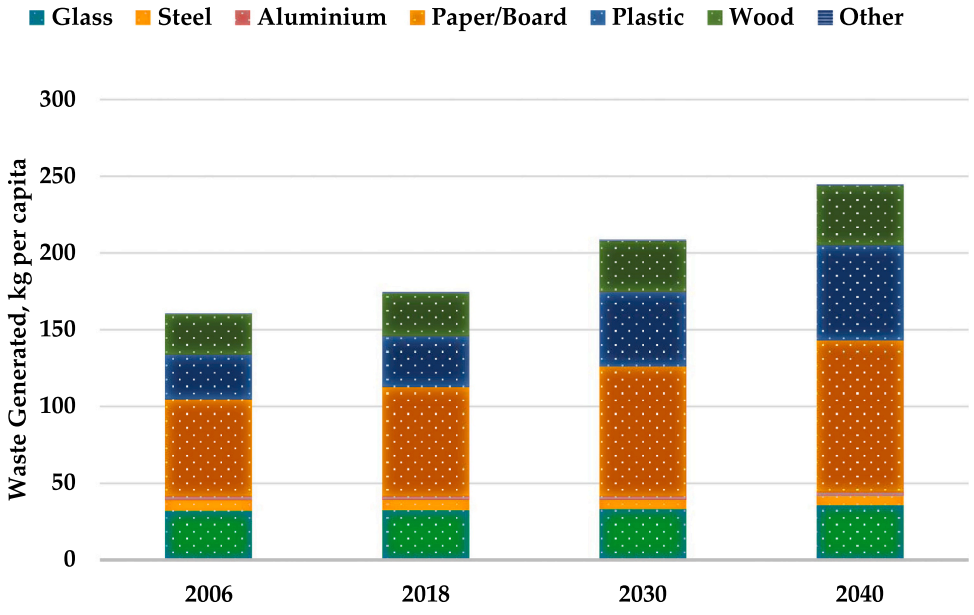


Fig. 1. The trend in packaging waste generation per capita (EU-27 countries).

processing that facilitate the integration of these materials into building projects. Growing attention to lifecycle assessments in construction and the trend toward net-zero carbon buildings will also stimulate the demand for sustainable materials (Anastasiades et al., 2021). Furthermore, the long history of technological advancements, evolving cultural standards, and environmental needs have influenced recycled and upcycled materials utilization in civil engineering and architecture. These materials will become more crucial to sustainable architecture in the years to come as the world struggles with urgent problems such as resource scarcity and climate change (González et al., 2021).

2.2. Strategies for incorporating recycled and upcycled materials in architectural designs

A rigorous strategy comprising material selection, design adaptation, collaboration with artists, and lifecycle assessment is essential when incorporating recycled and upcycled materials in architectural projects. Encouraging the use of recyclable and locally manufactured materials can substantially reduce the ecological impact of architects' work (Çimen, 2021). Design freedom makes it possible to incorporate these materials in distinctive ways, and collaboration with artists allows for innovative and unique applications that go against standard aesthetics. Architects can make choices that promote sustainability in the long term through their knowledge of material lifespan. By adopting these principles, the architectural industry can produce aesthetic and practical designs in constructing an environmentally friendly future (Kang et al., 2022).

2.2.1. Material selection

One of the most crucial steps in incorporating recycled and upcycled materials in architectural projects is careful material selection. Architects should use locally available materials with high recycling potential (Guerra and Leite, 2021b). Using recovered wood, recycled metal, and recycled brick reduces the need for virgin materials and minimizes transportation. Buying locally helps support locally owned companies and reduces carbon emissions compared to purchasing products over great distances. This approach satisfies ethical design standards and imparts uniqueness to a building since each salvaged material has its unique history (Christensen, 2021).

2.2.2. Design adaptability

Design flexibility is essential when using recycled and upcycled materials since the designs may not conform to standard specifications or criteria. Architects should produce designs that accommodate different shapes and sizes that allow the creative use of unconventional materials (Ghaffar et al., 2020). This adaptability may result in novel architectural solutions that defy traditionally accepted design principles. For example, an architect can use irregularly shaped repurposed wood in construction to produce unique designs that add aesthetical value while fulfilling functional objectives. Architects who embrace variety in design can create inventive and efficient techniques to use recyclable resources (O'Grady et al., 2021).

2.2.3. Collaboration with artists and craftsmen

One way to improve the utilization of reused materials in architectural projects is by collaborating with artists and artisans. These collaborations could lead to creative and novel use of materials, showcasing their potential in ways that would not be apparent in conventional construction (Eberhardt et al., 2022). Artists often have unique perspectives on material use and explore creative methods prioritizing sustainability and originality. For instance, a project can convert waste into a design's focal point using sculptural elements fabricated from recycled metal or glass. Such collaborations could empower builders and clients to value creative reuse and enhance knowledge of sustainable practices (Al-Saud et al., 2024).

2.2.4. Lifecycle assessment

Another essential element in the environmentally friendly design process is material life cycle assessment (LCA) (Adams et al., 2017). LCA evaluates the ecological effects of each stage of a material's lifecycle, from extraction and processing to use and disposal (Buyle et al., 2019). Architects who understand the long-term effects of the materials they use enhance the ecological viability of their projects. They may prioritize using certain materials, such as those with a high recycling rate or low embodied energy, over less sustainable ones (Dams et al., 2021a). LCA influences material selection and encourages architects to consider a building's whole life, thus leading to economically and environmentally sound designs (Akhimien et al., 2021).

LCA materials is a crucial component of environmentally friendly designs. It evaluates the environmental impacts of each stage of a material's lifecycle, from extraction and processing to use and disposal, and helps architects make more sustainable choices (Buyle et al., 2019). Architects can enhance the ecological viability of their projects by prioritizing materials with high recycling rates or low embodied energy. This approach influences material selection and encourages architects to consider a building's entire lifecycle in producing economically and environmentally sound designs that minimize waste and maximize efficiency, as illustrated in Fig. 2. Incorporating recycled and upcycled materials such as metal, wood, glass, concrete, plastic, and masonry reduces waste, energy consumption, and carbon emissions. Table 2 summarizes the integration of sustainability objectives across production, construction, utilization, recycling, and end-of-life phases to create a more resilient built environment.

3. Methodology

The methodology for conducting a literature review of recycled and upcycled materials in modern architecture and civil engineering focused on their uses, advantages, and challenges. The study analyzed the present research level in this field by setting defined goals, including identifying successful case studies, understanding the benefits and limitations of the materials, and analyzing creative methods in the industry. The eligibility criteria for this study are peer-reviewed papers and relevant case studies; it did not include non-English publications or research irrelevant to the architectural and civil engineering contexts (Manzoor et al., 2021). The literature search adopted a thorough technique across several academic databases: Scopus, Web of Science, and Google Scholar. This study used a defined set of keywords: "recycled materials in architecture" and "upcycled materials in civil engineering" and paired them with Boolean operators to narrow the results. Following the search, the papers were systematically screened for title and abstract reviews, followed by full-text assessments to confirm they match the inclusion requirements (Johnson and Onwuegbuzie, 2004). The study standardized the data extraction to obtain pertinent information, including the utilized materials, their uses, advantages, and challenges. The review also analyzed the quality of the studies using relevant assessment techniques to ensure the methodological rigor of the conclusions. The data was summarized qualitatively, emphasizing significant trends, patterns, and literature gaps. The summary will guide the debate concerning the significance of the results for practitioners and researchers and the ideas that encourage the use of recycled and upcycled materials in buildings. Finally, the evaluation will be published following the established rules, assuring openness and comprehensiveness in demonstrating the findings.

4. Research findings

According to recent research, recycled and upcycled materials in modern civil engineering and architecture have significant positive environmental and economic impacts by considerably reducing the requirement for virgin resources, preserving natural resources, and reducing the overall carbon footprint of construction projects (Pomponi

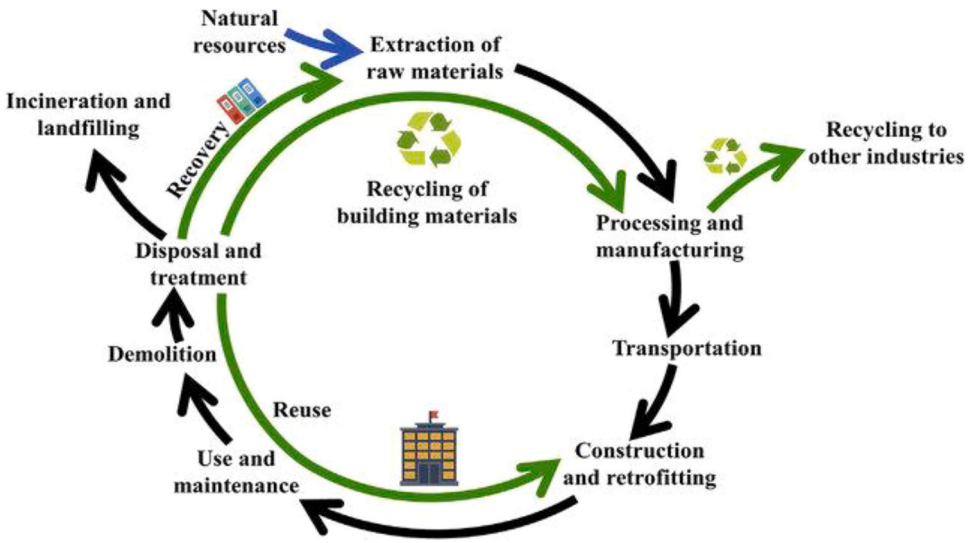


Fig. 2. Conceptual illustration of building material LCA (Huang et al., 2020).

Table 2
LCA phases for recycled and upcycled materials in contemporary architecture and civil engineering.

Material	Sustainability Objective	Materials and Production	Construction	Utilisation Phase	Recycling Phase	Upcycling Phase	End-of-Life Phase
Metal	Reduce mining, save energy, and reduce emissions	Recycled steel and aluminum from scrap and industrial waste	Easy incorporationinto structuralframeworks	High durability, corrosion resistance, and energy savings	Melted and reformed to produce new metal products	Reuse steel beams in new projects	Fully recyclable at high temperatures without loss of quality
Wood	Promote forest conservation and carbon sequestration	Reclaimed wood from old buildings and construction debris	Used in building frames, cladding, and flooring	Long lifespan if appropriately treated, renewable, and biophilic design	Shredded for fiberboard, mulch, or biofuel	Repurposed into furniture and smaller wooden structures	Natural decomposition if untreated or incinerated
Glass	Reduce sand mining, minimize energy consumption and emissions	Recycled glass from bottles, windows, and industrial waste	Incorporated into windows, facades, and decorative elements	High insulation properties and energy-efficient buildings	Crushed into cullet and melted to produce new glass products	Upcycled as decorative glass tiles and mosaics	Infinite recycling without loss of quality
Concrete	Reduce CO_2 emissions and the use of virgin aggregates	Recycled aggregates, fly ash, slag, and silica fume	Foundations, pavements, and precast elements	High compressive strength and durability and low maintenance	Crushed into recycled aggregates to produce new concrete	Concrete pieces reused in landscaping and barriers	Landfilling, if not properly recycled, and high embodied energy
Plastic	Reduce landfill waste and oil consumption	Recycled plastic from bottles, packaging, and industrial waste	Used in piping, insulation, and lightweight construction materials	UV resistance, flexible uses, and lightweight	Melted and remolded into new plastic products	Upcycled into innovative construction materials (e.g., eco-bricks)	Non-biodegradable, landfilled, or incinerated
Masonry	Reduce waste from demolition and promote a circular economy	Recycled bricks and masonry from demolition waste	Reused in walls, paving, and landscaping	High thermal mass and durability and minimal maintenance	Crushed for use as aggregates in new masonry	Repurposed for use in new buildings and artistic installations	Reused, repurposed, or crushed at the end of life

and Moncaster, 2017). Recycled steel and concrete, for example, have been shown to reduce the typically high greenhouse gas emissions (GHG) in conventional material production processes. Furthermore, recycled materials require less processing than newly manufactured components and thus consume less energy (Giesekam et al., 2014). From an economic perspective, recycled and upcycled materials are more affordable, notably if locally sourced, because of the lower processing and transportation costs. These factors make them a desirable substitute for cost-cutting measures and environmentally friendly building techniques in the construction industry (Mechouar et al., 2022).

There are challenges associated with the growing acceptance of recycled and upcycled materials. One of the primary concerns is the range in material performance and quality, which may raise doubt about their long-term reliability and safety in structural applications. Architects and engineers are ambivalent about using recycled and upcycled materials in large-scale projects (Adams et al., 2017). Regulatory constraints limit their use since current building practices and guidelines could make it difficult to use unconventional materials. Another challenge is market acceptance, where some stakeholders continue to doubt the reliability, robustness, and visual attractiveness of recycled and upcycled materials. It is essential to conduct more in-depth research to ensure material quality, update laws and regulations, and inform stakeholders about the benefits and potential of recycled and upcycled materials (Titirici et al., 2022).

Notwithstanding these drawbacks, the emerging innovative applications of repurposed and recycled materials demonstrate their viability and imaginative potential. Case studies from across the globe have illustrated the excellent performance of these materials when used in commercial and residential buildings. For instance, repurposing affordable and environmentally friendly shipping containers to

construct homes and commercial structures is gaining popularity due to their low cost and environmental friendliness (Sanchez and Haas, 2018). Similarly, repurposing metal, masonry, and wood in new constructions preserves their historical value and enhances the aesthetic appeal of modern buildings. Recycled glass has functional and aesthetic benefits when creatively used in building facades. These innovative projects are motivational examples of sustainable design, showcasing the versatility and potential of recycled and repurposed materials (Dams et al., 2021).

These findings suggest that, with further innovation, collaboration, and research, recycled and upcycled materials can strongly influence the direction of sustainable architecture and civil engineering. Ongoing initiatives are necessary to address the problems related to using these materials, including improving quality control, modernizing laws, and increasing market acceptance to encourage their widespread adoption (Chan et al., 2020). Recycled and upcycled materials will probably increase in significance with the advancements in the construction sector to create more resource-efficient, environmentally friendly, and visually pleasing built environments (Saieg et al., 2018). Engineers and architects can use these techniques to develop a more responsible and sustainable construction philosophy that will benefit society and the environment (Halliday, 2008).

4.1. Sustainable development

Recycling and upcycling are increasingly used by civil engineers and architects to advance sustainable development by minimizing the use of virgin resources, reducing waste, and supporting environmental objectives (Zhang et al., 2023). As illustrated in Fig. 3, raw materials are initially used to produce items like seatbelts. Once these products reach the end of their lifecycle, they are often discarded and sent to a junkyard. Recycling processes break these materials down into reusable fibers, while upcycling collects and transforms them into higher-value items, such as bags. These practices not only reduce waste and extend material lifecycles but also promote a circular economy, where objects are repurposed for different uses rather than being discarded, ultimately safeguarding natural resources. However, in terms of economics, using recycled and upcycled products reduces expenses and supports locally-owned businesses. They are more cost-effective since the materials can be locally sourced and require less processing, thus making the buildings more affordable (Allwood, 2014). Apart from creating new markets and sectors, the growing demand for these materials encourages technical developments in the construction and material processing sectors. Using recycled and upcycled materials allows the construction industry to build a more resilient economy by reducing reliance on virgin resources and increasing flexibility in responding to changing environmental conditions and market developments. Buildings



Fig. 3. Primary benefits of material recycling and upcycling.

constructed using recycled and reused materials promote social inclusion and community participation. Local sourcing of the materials encourages community participation in projects for sustainable development and strengthens regional economies (Ljungberg, 2007). It is possible to use these materials in construction by reflecting and preserving cultural legacies, thus enhancing the buildings' social and cultural relevance. Incorporating recycled and upcycled materials in public buildings serves as teaching examples for sustainable endeavors and increases awareness of ethical usage and manufacturing. Recycling and repurposing materials in civil engineering and architecture help to promote social inclusion, economic resilience, environmentally friendly practices, a fairer and more sustainable built environment, and environmental preservation. As the construction industry grows, these approaches will become increasingly crucial because they enable a future where growth fulfills the needs of the present and next generations (Allwood et al., 2011; Moshood et al., 2024).

4.2. Circular economy

The circular economy in modern architecture and civil engineering, as illustrated in Fig. 4, focuses on the continuous use of resources through material recycling and upcycling, in contrast to the traditional linear model of "take, make, dispose." This approach significantly reduces waste by integrating recycled materials like concrete, steel, and glass into new construction projects (Condotta and Zatta, 2021). Architects and engineers contribute to preserving natural resources, promoting sustainable material usage, and minimizing the environmental impact of building operations. A key component of the circular economy is designing infrastructure with longer lifespans, adaptability, and ease of disassembly, which allows materials to be collected and reused after the lifespan of a building (Leising et al., 2018). Preparing for future resource reuse enables the construction industry to close the loop and optimize resource efficiency. The circular economy reduces material costs and fosters innovation in construction methods since locally sourced recycled and upcycled materials are more affordable than new ones, which helps boost regional economies (Assefa and Ambler, 2017). Circular building techniques encourage advancements in material science and design while creating jobs and stimulating the economy. Socially, they promote fairer resource distribution and provide access to sustainable infrastructure and housing. As the construction industry increasingly adopts circular economy principles, recycled and upcycled materials will play a more significant role in sustainable building practices. Collaboration among various sectors will be essential in developing the required regulations, technologies, and educational programs. Looking ahead, civil engineers and architects will focus on designing structures for disassembly, utilizing reusable materials, and embracing the circular economy to build a more resilient and sustainable environment (Arslan, 2007; Cottafava and Ritzen, 2021).

The recycling rates of materials in the European Union varied significantly. In 2022, the Netherlands had the highest recycling rate of 27.5 %, whereas Finland recorded the lowest of only 0.6 %. On average, 11.5 % of the materials used in the EU were recycled products, a notable increase from 2020, when the average recycling rate was 8.2 %. The projected recycling rate for 2025 is even higher, and this is encouraging news for the environment since it means reduced resource consumption, lower emissions, and less waste disposed of in landfills.

4.3. The 3R

In contemporary architecture and civil engineering, the 3R principle (Reduce, Reuse, Recycle) provides the fundamental framework that encourages sustainability, especially when using recycled and upcycled materials (Honic et al., 2019). Fig. 5 compares the circular and linear economies, highlighting the flow of materials. In the circular model, resources are optimized: the design phase focuses on sustainability based on R-strategies (R2: Reduce), the consumption phase ensures

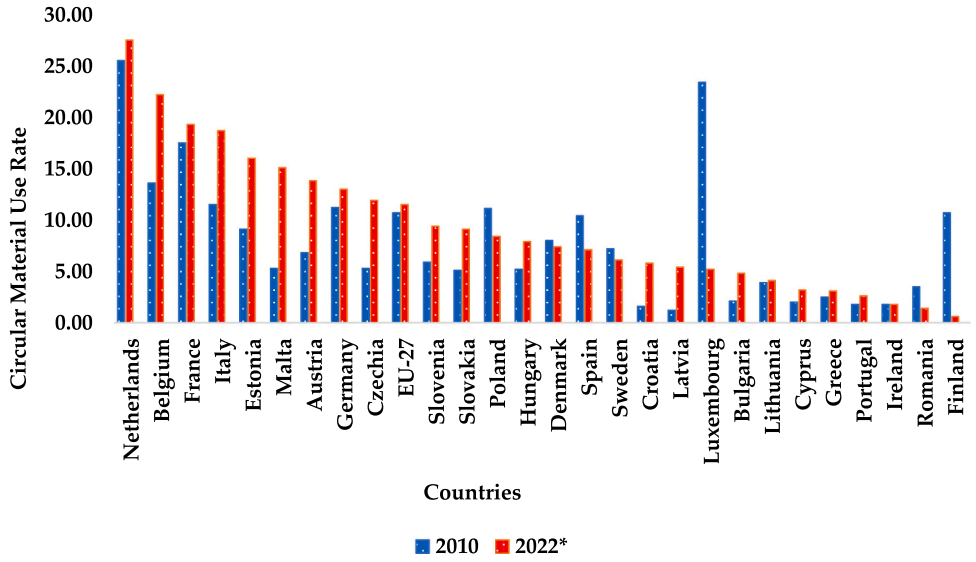


Fig. 4. The waste material transformation process to achieve a circular economy.

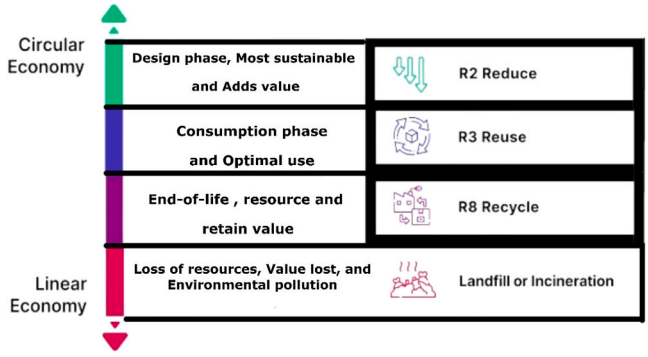


Fig. 5. Reduce, Reuse, Recycle based on R strategies (adapted from Larae and Daphne, 2023).

optimal use (R3: Reuse), and the end-of-life phase retains value through recycling (R8: Recycle). In contrast, the linear model leads to resource loss, pollution, and waste disposal via landfills or incineration. The circular approach treats waste as a resource, promoting regional sustainability through efficient waste management, reduced material use, and recycling. This strategy fosters innovative design, supports sustainability goals, and minimizes environmental impacts. The more sustainable built environment resulting from implementing 3R will encourage the adoption of more environmentally friendly building methods and promote the broader objectives of sustainable development and circular economy (Victar et al., 2024).

4.3.1. *Reduce*

The first component of 3R, reduce, focuses on reducing resource consumption and waste generation at the source. Civil engineers and architects achieve this objective by adopting intentional design methods that help achieve efficiency and sustainability (Van Gulck and Steeman, 2024). For example, architects can reduce the need for new resources by designing buildings that maximize energy efficiency or use fewer materials. Using robust and premium materials can extend the lifespan of structures and reduce the need for restoration and the resulting waste generation. The emphasis on reduction is to encourage a proactive attitude towards ecological responsibility and minimize the environmental impact of construction projects (Phonthanukitithaworn et al., 2024).

4.3.2. *Reuse*

The second component, reuse, emphasizes the significance of recycling components and materials whenever possible. Reusing is particularly relevant given how often modern architecture deconstructs and refurbishes. Instead of demolishing old structures and disposing of their components in landfills, architects and engineers can carefully deconstruct them to recover materials for new projects. This approach preserves resources and the embodied energy of the reused materials (Oyejobi et al., 2024; Zorpas, 2024). Reusing may include the upcycling process that transforms waste materials into new, more valuable products. One way to extend a material's lifespan while reducing waste is reusing the wood salvaged from historic buildings to make furniture or decorative elements (Erol and Thöming, 2005; Rose and Stegemann, 2019).

4.3.3. *Recycle*

The final component of 3R, recycling, involves restoring materials to their original condition of use. In architecture and civil engineering, recycling means recovering resources from construction and demolition sites, such as concrete, metals, and glass (Galila et al., 2024), and processing and converting them into new construction materials to reduce the need for virgin resources. Recycled concrete can be used as aggregates in new concrete mixtures, and metals can be melted and used for many purposes. Recycling seeks to prevent waste from ending in landfills and promote development (Nilimaa, 2023).

4.4. *Recycled and upcycled materials in architectural designs*

Incorporating recycled and upcycled materials in architectural projects has significantly transformed the construction industry's commitment to sustainability and environmental responsibility (Almusaed et al., 2024). Recycling and upcycling enable the creation of functional, aesthetically pleasing, and environmentally conscious structures from diverse materials, including metal, wood, masonry, glass, concrete, and plastic. Each material has unique qualities and applications that support innovative design concepts and promote resource efficiency. By creatively repurposing the materials that would otherwise contribute to waste accumulation, architects reduce landfill burden, conserve resources, reduce building carbon footprints, and inspire novel design ideas. This strategy aligns with broader sustainability objectives and principles of a circular economy that emphasize reusing resources for as long as possible. Table 3 summarizes the findings from previous case studies and real-world examples and highlights the potential of recycled

Table 3
Perspective on the potential of recycled and upcycled materials in construction and civil engineering.

Material	Country	Case study	Application	Benefits	Challenges	Authors/ Year
Metal	USA	Recycled steel in skyscrapers	Structural support in tall buildings	High strength, durability, lower CO_2 emissions	High energy consumption during the recycling process	(Finnigan, 2019)
	Germany	Cladding for modern buildings	Aluminum facades from reclaimed sources	Lightweight, corrosion resistance	Limited availability of reclaimed aluminum	(Ivanica et al., 2022)
	Japan	Recycled steel for bridges	Bridge construction	Cost savings, sustainability	Quality control during recycling	(Kitada, 2006)
Wood	USA	Reclaimed timber for residential homes	Structural and aesthetic components	Natural insulation, reduced deforestation	Limited supply of high-quality reclaimed wood	(Bergsagel and Heisel, 2023)
	Italy	Upcycled wood pallets for interior designs	Interior furnishings and partitions	Low cost, aesthetic appeal	Potential issues with durability and insect infestation	(Altamura and Baiani, 2019)
	Türkiye	Reclaimed wood in urban housing projects	Low-cost housing construction	Sustainable, cost-effective	Moisture retention causes decay	(Işik, 2003)
Glass	Australia	Recycled glass in curtain walls	Facade of commercial buildings	Energy-efficient, aesthetic value	Brittleness and thermal performance	(Cheong et al., 2024)
	USA	Glass cullet in concrete production	Aggregate replacement in concrete	Reduce the demand for raw materials, waste diversion	Compatibility with concrete mixtures	(Tucker et al., 2018)
	Spain	Recycled glass tiles in flooring	Interior and exterior flooring	Aesthetic value, durability	Limited mechanical strength	(Gabaldón-Estevan et al., 2014)
Concrete	Spain	Recycled concrete aggregates in road bases	Road construction	Resource conservation, lower cost	Inferior mechanical properties compared to virgin aggregates	(Lopez-Uceda et al., 2020)
	Spain	Recycled concrete in building foundations	Foundations of residential and commercial buildings	Reduce waste, lower cost	Quality assurance and varying strength	(Ortiz-Lozano et al., 2022)
	India	Concrete with fly ash additives	Reinforced concrete structures	Higher durability, use less cement	Workability challenges	(Nayak et al., 2022)
Plastic	South Africa	Recycled plastic lumber for outdoor decks	Outdoor decks and furniture	Durability, low maintenance	UV degradation over time	(Tenele Unity Dlamini, 2020)
	Ecuador	Recycled PET in construction blocks	Alternative to traditional masonry	Lightweight, sustainable	Limited structural performance	(Genovez et al., 2021)
	Switzerland	Recycled plastic insulation in roofs	Roof insulation	Thermal performance, low cost	Fire resistance oncerns	(Superti et al., 2021)
Masonry	Algeria	Recycled brick in historical building restoration	Restoration projects	Preservesheritage, cost savings	Limited availability of matching bricks	(Ayat et al., 2022)
	Spain	Recycled concrete masonry units (CMUs)	Load-bearingwalls incommercialbuildings	Durability,sustainability	Lower strength compared to virgin CMUs	(Madrid et al., 2017)
	Spain	Crushed masonry in road sub-bases	Road construction	Resourcereuse, cost-effective	Quality and consistency issues	(Agrela et al., 2012)

and upcycled materials in construction and civil engineering. It is worth noting that these studies also underscore the need for further research to gain better insight into the challenges and limitations associated with utilizing these materials.

4.4.1. Metal

The manufacture of metal products, from ore extraction to product manufacturing, is energy intensive. However, metal has the most potential for further processing, remelting, and other treatments (Furlani et al., 2010). Crucially, repeating this process can extend the life cycle of raw materials. The proof of metals' worth is that they are recovered after demolitions and seldom disposed of in landfills. Metal components are often reconditioned since they can be disassembled and used in different constructions. In this regard, steel structural components with standardized cross-sections are the most likely to be reused (Karadağ and Aslan, 2018). One example is a concert hall in Vienna, Austria, constructed with copper cladding. Metal recycling is energy-efficient since it consumes less energy than mining and manufacturing new metal products. Examples of structural components produced from recycled steel are beams, columns, and reinforcements. Apart from conventional steel, aluminum is another recyclable metal often used in construction. Its lightweight, strong, and corrosion-resistant properties render it ideal for various applications, including windows, roofing, and façade (Marriage and Gjerde, 2024). Architects can prevent waste and provide unique architectural components by upcycling metal, such as abandoned shipping containers or recovered metal panels. Generally, recycled metal strengthens a structure and extends its lifetime while conserving resources (Ling et al., 2024).

4.4.2. Wood

Wood, a common component in construction, is easy to recycle and upcycle. Reclaimed wood from old homes, barns, or furniture enhances architectural designs due to its unique character and historical significance. The natural beauty of oak makes it a popular choice for beams, flooring, and accent pieces (Piccardo and Hughes, 2022). Among the ways to recycle scrap wood, which otherwise would be wasted, is turning it into engineered goods like laminated veneer lumber (LVL) and oriented strand board (OSB), which have excellent structural performance (Garcia et al., 2024). Furthermore, using responsibly harvested wood in construction reduces the need for virgin timber and encourages

ethical forestry methods. It also has significant environmental advantages, such as reduced carbon emissions and better energy efficiency when constructed with a lifetime in mind (Garcia et al., 2024).

4.4.3. Masonry

Masonry components from demolished structures, such as bricks and stones, can be repurposed or recovered. Recycled masonry has a distinctive visual appeal and offers a sustainable substitute for new materials. Reclaimed bricks, for instance, can be used as wall cladding, landscape accents, or even structural elements to impart personality to conventional and modern designs (Keshmiry et al., 2024). A common way to recycle masonry is by crushing and reusing masonry debris as aggregates for fresh concrete or filler. It preserves natural resources by reducing the demand for new aggregates and diverting waste from landfills. Including recycled masonry components in designs improves sustainability and demonstrates the commitment to conserving the craftsmanship and heritage of ancient buildings (Kubica, 2024).

4.4.4. Glass

Glass is often utilized in buildings as single-glazing panes for interior doors, multi-chambered windows, elevation panels, and glass railings. Recovering glass in its original state and repurposing it in another facility is an innovative and eco-efficient approach. However, disassembling and transporting glass is challenging because of the risk of damaging the delicate parts. Additionally, the specifications for these parts may be an issue, particularly the thermal insulation qualities of glass packages, which often fall short of the current construction regulations (Rincón et al., 2016). Recycled glass is an adaptable material suitable for numerous architectural projects. Glass recycling involves collecting, cleaning, and melting discarded glass to produce new glass products (Vefago and Avellaneda, 2013). This process consumes less energy than manufacturing glass from raw materials, making it an ecologically responsible choice. Recycled glass is suitable for, among others, tiles, countertops, and insulating materials (Jiang et al., 2019). It can also be integrated into building facades as beautiful components that add energy efficiency through insulating glass modules. Upcycling glass includes artistically recycling bottles and jars into unique design features or ornamental components in indoor and outdoor environments. The aesthetic flexibility of recycled glass helps architects increase the visual appeal of their designs while promoting sustainable practices (Kazmi et al., 2021).

4.4.5. Concrete

Concrete is the primary component of buildings. Of the materials used in the recently constructed structures, the highest proportion is the combination of cement, water, aggregates, reinforced concrete reinforcing steel, and chemical additives (Meyer, 2009), despite the materials causing strain on the natural surroundings. The environment is influenced primarily by how natural resources are used and the costs of their extraction, processing, and transportation (Makul, 2020; Meyer, 2009). Concrete components generally end as waste after the end life of the building (Petek Gursel et al., 2014). Since there is no way to recover the individual components in concrete, it has a low value as a demolition material and is considered demolition debris. However, it is possible to turn broken-down concrete leftovers from remodeling or demolition operations into recycled concrete aggregates (RCA) (Ibrahim et al., 2023). Using them in fresh concrete mixes reduces the need for virgin aggregates and preserves natural resources. Apart from recycling old concrete, upcycling concrete includes ingeniously reusing concrete components, such as using precast concrete panels or furnishings. Concrete's resilience makes it possible to shape it into architectural installations, landscape pieces, or ornamental embellishments. Architects can make their projects more sustainable without sacrificing a structure's life span and structural integrity by integrating recycled and repurposed concrete into their designs (Sieffert et al., 2014).

4.4.6. Plastics

Polymers, such as PVC, HDPE, polypropylene, and polystyrene, are used in construction as window frames, pipes, foams, and insulation. Depending on their components, plastic materials can be recycled into various items, such as window frames, insulation, carpets, tiles, interior linings, synthetic coating, and installation components (Agarwal and Gupta, 2017). The increasing use of recycled plastic in contemporary civil engineering and architecture is due to its adaptability and lightweight properties (Cudzik and Kropisz, 2024). It is possible to process and turn various plastic waste, such as bottles, containers, and packaging, into new construction materials. Recycled plastic timber is suitable for decking, fencing, and outdoor furniture since it is an environmentally beneficial substitute for conventional wood items (Real, 2023). An example of incorporating recycled plastics artistically into architectural projects is using them as ornamental components or panels for facades. A creative way to use plastic waste is by recycling it into insulation, tiles, and structural components. Recycling plastics encourages resource economy and environmental responsibility in construction methods while reducing the plastic waste disposed of in landfills (Satti et al., 2024).

4.5. Case studies and real-world examples

This section presents relevant case studies and real-world architectural examples to illustrate the many uses of recycled and upcycled materials to build artistically beautiful and functional structures. The illustrations from homes and municipal parks demonstrate how imaginative designs can transform waste into valuable resources and create a more sustainable built environment. The future of architecture will be more resource-efficient and environmentally conscientious if architects continuously look for potentially recyclable materials (Zhang et al., 2015; Y. Zhang et al., 2022).

4.5.1. The brighton waste house

Architect Duncan Baker-Brown designed the Brighton Waste House at the University of Brighton in the United Kingdom, a superb example of a creative ecological structure that heavily uses recycled and upcycled materials (Bagaeen, 2014). This experimental project is notable for its commitment to sustainability since it uses over 85 % recycled materials. The unusual relics used to construct the house include old denim pants as insulation, discarded toothbrushes, and VHS tapes (Duncan, 2016). The design emphasizes repurposed waste materials to build practical homes while addressing grave environmental issues. The waste house is a functional home and educational resource constructed to raise community awareness of sustainability and waste reduction (Bagaeen, 2014). The program involved experts, students, and the public in seminars and educational tours to discuss the benefits of recycling and upcycling in its construction. This endeavor encourages the adoption of similar strategies in future projects and advances a more comprehensive understanding of sustainable building concepts. The success of the Brighton Waste House illustrates how to integrate waste materials into architectural design without sacrificing aesthetics or quality. The inventive use of leftover materials demonstrates how ingenuity can transform waste into valuable resources and push the boundaries of conventional construction techniques (Bagaeen, 2014; Durrant et al., 2018).

4.5.2. The Bullitt Center

Built in 2013 and regarded as one of the greenest commercial buildings in the world, the Bullitt Center in Seattle showcases sustainable architecture (Homchick Crowe, 2020). The Bullitt Foundation commissions the project and hires the Miller Hull Partnership to design the building. The design meets the strict criteria for the Living Building Challenge and embodies sustainability via water self-sufficiency, net-zero energy consumption, and the utilization of non-hazardous, ethically produced materials. Based on the long life, loose fit concept, the

design of the Bullitt Center ensures its adaptability and lifetime of at least 250 years while reducing the ecological impact of frequent repairs or demolitions (Homchick Crowe, 2020; Peña et al., 2017). The extensive use of recycled and repurposed materials significantly enhances the Bullitt Center's environmental sustainability. Using recycled steel as a substitute for new steel to fabricate vital structural components reduces the carbon footprint of its construction. The reclaimed wood collected from old warehouses and barns to make furniture and flooring preserves natural resources while adding to the interior character. Salvaged glass was recycled to make the windows and skylights to maximize natural light and reduce the need for artificial lighting, thus enhancing energy efficiency during construction. The Bullitt Center's dedication to sustainability was rewarded with the LEED Platinum certification and recognition as the first commercial office building to meet the Living Building Challenge. Besides its physical shape, the Bullitt Center is a living laboratory and teaching tool that motivates the construction industry to adopt greener construction methods. Its fresh approach to using recycled and upcycled materials establishes a new benchmark for sustainable urban development by proving that high-performance buildings can be environmentally responsible and aesthetically pleasing (Peña et al., 2017).

4.5.3. The silo in Copenhagen

An iconic example illustrating the prospects for adaptive reuse in sustainable urban development is the Silo in Copenhagen. Originally constructed as a grain storage facility in the 1960s, COBE Architects modified The Silo in 2017 and turned it into a modern residential and mixed-use building. The architects retain the original concrete construction during the modification to honor the industrial past of the building while fulfilling the current needs. This approach conforms with sustainable development ideas in mitigating the impacts of new construction and demolition on the surroundings (Zamojski, 2023). The renovation of Silo uses primarily recycled and reused materials. The aluminum panels for the new exterior enhance its modern appearance and contribute to energy economics. The salvaged wood imparts character and warmth to the room, while the interior flooring, staircases, and other elements are from steel. The selected materials reflect the project's commitment to sustainability by repurposing current materials to create aesthetically pleasing environments. The Silo shows the significance of using recycled materials and conserving historic buildings during urban regeneration; it represents sustainable architecture in Copenhagen. The project illustrates the possibility of reasonably modifying industrial buildings to fulfill modern demands while mitigating environmental impacts. The distinctive design and environmentally friendly approach of the Silo earned it recognition as a model for future projects, inspiring others to mix historical conservation with sustainable urban development (Grindsted et al., 2024).

4.5.4. The Center for Sustainable Energy Technologies (CSET)

The Ningbo, China-based Center for Sustainable Energy Technologies (CSET) is an example of integrating recycled materials into educational architecture (Cao et al., 2022; Deng and Cheshmehzangi, 2018). The reclaimed bricks for its facade provide insulation and esthetic beauty. CSET encourages utilizing renewable energy resources for environmentally friendly living by harnessing solar and wind power. Using recycled materials conforms with the center's objective to educate the public on sustainability and energy economics. This project provides an example of using recycled materials in educational settings to promote sustainability. The creative design and construction techniques of CSET will benefit future generations fascinated with sustainable architecture (Deng and Cheshmehzangi, 2018; Guo et al., 2020).

4.6. Opportunities and challenges in using recycled and upcycled materials

Modern civil engineering and architecture must deal with the formidable challenges and fascinating opportunities of using recycled and upcycled materials. Despite the many barriers preventing their widespread use, these materials provide a pathway toward sustainability as the construction industry seeks to mitigate environmental impact. It is essential to have a sound understanding of lifecycle analysis, technological advancements, and economic viability to seize the opportunities and overcome these challenges to fully exploit the potential of recycled and upcycled materials and contribute to a more durable, ecologically friendly, and efficiently built environment (Malabi Eberhardt et al., 2021).

4.6.1. Technological advancements

The primary challenges in using recycled and upcycled materials are ensuring the quality and uniformity of the materials (Dogu et al., 2021; Krishnan et al., 2021). Unlike virgin materials, recycled and upcycled materials often have different compositions and performances since they have been used previously and have undergone recycling (Krishnan et al., 2021). Recycled steel may have leftover components from previous usage that compromise its structural integrity. Similarly, recycled concrete may contain impurities or have lower compressive strength. These discrepancies impede the fulfillment of the stringent requirements for large-scale building projects (Gencel et al., 2012; Kirchherr et al., 2017).

Innovative processing and manufacturing techniques are crucial to ensure effective waste material processing and recycling into high-quality construction products. These technologies are not yet available for some materials, such as complex composites and mixed polymers. There are difficulties in developing and scaling these technologies to meet the requirements of large-scale building projects. The industry has to develop economical methods to process waste materials to ensure they can be used consistently and safely in construction without sacrificing quality (Kirchherr et al., 2017)

Despite these hindrances, there are plenty of opportunities for innovations in recycling technologies. The quality and usefulness of recycled materials will likely improve with more in-depth research on novel recycling strategies, such as chemical recycling and complicated processing. It is possible to manufacture more durable, longer-lasting composite materials by adding performance-enhancing chemicals to the recycled materials. Advancements in concrete technology that incorporate cementitious additives into recycled aggregates could produce high-performance and environmentally friendly concrete (Virtanen et al., 2017).

Concerning material enhancements, new technologies can facilitate the creation of high-performance composite materials comprising recycled elements and additives. For example, adding fibers or fillers to recycled plastics can produce materials with higher strength and durability than standard plastics. Similarly, advances in concrete technology allow the incorporation of supplementary materials into recycled aggregates to obtain concrete with enhanced properties and less adverse environmental impact. These innovations highlight how recycled resources can support ecologically responsible, high-performance construction (Ko et al., 2024).

4.6.2. Lifecycle analysis

Lifecycle analysis (LCA) is a vital tool for assessing the ecological effects of materials at each stage of their life cycle, from extraction to disposal. However, it requires accounting for many factors, such as the energy consumption and emissions released during the procurement, processing, and delivery of the recycled materials and their maintenance and long-term functionality. The lack of established methods and reliable data in the construction industry makes it difficult to conduct accurate LCAs (Li et al., 2019)

The inherent complexities of lifecycle assessments of the materials fabricated from recycled materials present serious challenges, including standardizing the assessments because, unlike virgin materials, recycled materials have varying compositions and histories. Moreover,

assessments of the overall impact are vague since the energy required for the recycling process occasionally surpasses the environmental benefits. It can be challenging to ascertain the actual sustainability benefits of some upcycling techniques since they involve treatments or interventions that raise new environmental concerns (Craighill and Powell, 1996). It may be incomprehensible how using recycled materials would benefit the environment since its impacts are unknown. Despite the perception that recycling benefits the environment, the energy expense of a recycling process may outweigh some of its advantages. For instance, there may not be a clear net environmental advantage when recycling some materials since the energy consumption may equal that of manufacturing new materials. Furthermore, even though upcycling adds value to waste materials, it may occasionally involve chemical treatments or other methods with adverse environmental consequences, thus confounding their sustainability evaluation (Birat, 2015).

Standardized procedures and advancements in LCA technologies can provide ways to deal with its challenges. Architects and engineers can make educated judgments of the environmental impacts of using recycled materials if they have access to accurate data. The construction industry can improve its decision-making process and resource efficiency by quantifying the sustainability of recycled materials through a simplified and standardized LCA (Chiu et al., 2008).

Holistic sustainability strategies like LCA are crucial in creating comprehensive environmentally friendly approaches that examine a material's short- and long-term implications. By adopting this approach, the construction industry can opt for more ecologically friendly alternatives and ensure that recycled materials provide immediate benefits and contribute to achieving long-term sustainability goals (Gaines, 2012).

In southern Italy, Colangelo et al. (2018) conducted an LCA to evaluate the environmental impacts of recycled concrete mixtures. They compared three mixtures: the first mixture was incorporated with construction and demolition (C&D) waste, the second was incorporated with marble sludge, and the third was added with cement kiln dust (CKD) (Hasheminezhad et al., 2024). The researchers used the SimaPro software and Eco-Indicator 99 method to analyze 37 scenarios, including variations in recycling distances. The results revealed that the mixtures containing C&D waste and CKD had the lowest environmental impact. Fig. 6 illustrates how, in the short term, recycling significantly reduced GHG and other environmental impacts, thus proving the environmental and potential financial benefits of recycled concrete (Colangelo et al., 2018).

In 2018, (Hossain et al., 2018) conducted LCA to evaluate the feasibility of upcycling wood waste into cement-bonded particleboard (CBP) across four scenarios. They found that the wood waste recycling rates in Hong Kong resulted in significantly lower GHG emissions, thus underscoring the benefits of technological innovation in managing wood waste disposal and improving sustainability in construction. Table 4 shows that cement binder was the primary contributor to GHG (over 80 %), while the contribution of wood particles and additives to GHG was much lower. The study concluded that it is environmentally beneficial and cost-effective to upcycle wood waste into CBPs because it provides a sustainable solution for eco-friendly construction materials.

4.6.3. *Economic viability*

There are several obstacles to ensuring the economic viability of using recycled and upcycled materials in construction. In particular, when working with complex waste streams, the costs associated with gathering, classifying, and processing waste may be higher than the incurred costs when using new resources. The varying prices and demand for recovered materials in the market add to the financial burden and could potentially undermine the stability and profitability of recycling operations. These are among the obstacles to the economic feasibility of using recycled materials in construction that constrain their widespread use (Duran et al., 2006).

The high cost of recycling and upcycling impedes the advocacy for recycled and upcycled materials. The complex procedures may make waste acquisition, sorting, and recycling more expensive than creating new materials. This issue is particularly evident in managing complex waste streams since it requires additional sorting and purification processes. The inconsistent market for recycled materials, where the demand and price fluctuate with overall economic conditions, further exacerbates the financial consequences of recycling activities (D'Adamo

Table 4
GHG emissions (%) from CBP production (Hossain et al., 2018).

Sector	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Wood aggregate	4.12	2.96	3.25	3.38
Binder	55.99	65.51	71.87	74.59
Accelerator	1.06	-	-	-
Production process	32.8	31.53	34.59	35.89
Transport to Hong Kong	6.03	-	-	-
Carbon uptake	-	-	−9.71	−10.08
Avoided impact	-	-	-	−3.78
Total (%)	100	100	100	100

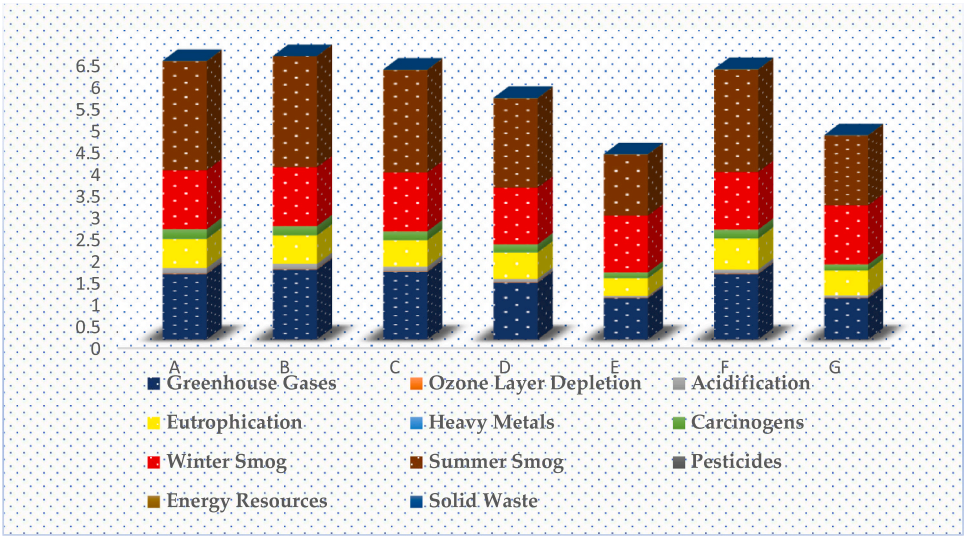


Fig. 6. Comparison of the consequences of disposing of concrete containing natural aggregates versus C&D aggregates (Colangelo et al., 2018).

et al., 2017).

The marketability of recycled materials hinders their economic viability. Concern for recycled product's appearance, functionality, and quality often prevents acceptance. Recycled materials may not be as popular or accepted in mainstream construction projects since developers and consumers view them as inferior. This perception and the lack of popular demand for recycled materials may hamper the establishment of economies of scale that will increase their price and reduce the competitiveness of recycled materials relative to traditional solutions (Duran et al., 2006).

Economic incentives, public policy, and legislative support can help address challenges and increase the viability of recycled materials from an economic perspective. Grants, tax credits, and other financial incentives for adopting sustainable construction practices can help offset the higher cost associated with recycling. In addition, laws mandating or encouraging the use of recycled materials in construction projects can ensure a steady market, increase demand, and spark interest in recycling technologies, thus making the recycled products business more competitive and productive and ultimately leading to lower prices and better revenue opportunities (Zhao et al., 2010).

In the long term, the cost savings and benefits of using recycled products may outweigh the initial high cost. Buildings constructed using recycled and repurposed materials are more durable, energy efficient, and require less care, and these benefits add up to financial profits throughout their lifespan. Apart from their competitive advantage over conventional materials, recycled materials become more inexpensive upon reaching economies of scale and advanced recycling methods. In summary, recycled materials may be a more financially viable option and provide significant long-term benefits to the construction industry (Marques et al., 2014).

5. Conclusion

Modern architecture and civil engineering focus on the vital role of repurposed and recycled materials in fostering environmental awareness and a sustainable built environment. Repurposed and recycled materials offer a practicable approach to reducing carbon emissions and protecting natural resources, which are increasingly becoming a grave concern in the construction industry. Using repurposed and recycled materials reduced waste and mitigated the environmental impacts of standard construction techniques, thus minimizing the demand for virgin resources. The efficient use of repurposed and recycled materials in well-known structures demonstrates how well these materials meet the structural, aesthetic, and performance requirements of modern civil engineering and architecture. Besides upcycled wood, glass, and plastic, researchers have illustrated the value and adaptability of recycled metals and concrete in constructing imaginative, durable, and ecologically responsible buildings. These approaches also promote a circular economy by extending the lifespan of materials and mitigating the environmental impacts of construction activities. They have demonstrated the possibility of designing aesthetically pleasing, functional, and high-performing buildings through skillful use of recycled and repurposed materials in well-known projects like the Bullitt Centre in Seattle and The Silo in Copenhagen. The Bullitt Centre and Silo demonstrate how recycled materials offer distinctive design choices, enhance the structures' durability and efficacy, and illustrate the synergistic link between sustainability and creativity. It is imperative to increase and expand the use of recycled and upcycled materials in the future to address global concerns about sustainability and climate change. Recycled and upcycled materials will shape the construction industry's future as more architects, engineers, and developers utilize them. The shift toward adopting more sustainable approaches in construction, which support environmental goals and foster innovation and progress, ushers in a new era of thoughtful and sound design and civil engineering.

5.1. Recommendations for future studies

- Future research to develop new recycling technologies should focus on improving recycling methods to enhance the merit and practicality of using recycled building components. They should identify effective techniques for recycling difficult-to-manage materials like composites and mixed-material wastes.
- Future researchers should evaluate recycled and upcycled materials across their entire lifecycle by conducting lifecycle analysis and environmental impact assessment to gain better insight into their long-term impacts relative to traditional materials. They should also estimate resource exhaustion, energy consumption, and carbon emissions throughout the materials' lifespan.
- Evaluate the effects of different climates and circumstances on the efficacy of using repurposed and recycled materials in varying climates and construction methods. Understanding how they behave in varying conditions, such as extreme heat, high humidity, or seismic zones, will facilitate the widespread applications of these materials and ensure their long lifespan and safety.
- Investigation of adopting a circular economy should focus on the building design process, material repurposing, and resource recovery to ensure its sound adoption in the construction industry. The investigations should aim to convince engineers and architects to design structures that maximize material reuse at the end of their lifetime.
- Economic research should perform cost-benefit analyses of integrating recycled and upcycled materials in construction projects to gain better insight into the financial benefits of sustainable construction practices. They should consider the upfront expenses, long-term savings, and potential economic incentives, including tax breaks and subsidies.
- Investigate the social and cultural impacts of using recycled and upcycled materials in civil engineering and architecture on society and culture. These investigations should assess the consensus, acceptance, and cultural significance of incorporating recycled and upcycled materials into built environments, particularly in historically or socially significant locations.
- Examine the role of legislative and administrative bodies in formulating policies and regulatory frameworks that promote the use of recycled and upcycled materials. Researchers should also study the outcomes of the best practices adopted in other locations and propose new regulations or incentives. In addition, they should also evaluate how well the present policies promote sustainable construction practices.
- Investigate novel design, aesthetic potential, and innovative design opportunities in using repurposed and recycled materials. The investigations should explore the methods for using recycled and upcycled materials to produce unique visual effects and integrate them into innovative architectural designs that push the boundaries of sustainability and innovation.
- Study the impact of educational and awareness campaigns to achieve better outcomes in using repurposed and recycled materials. The studies should assess the performance of the education programs targeting engineers, architects, and the general public and propose strategies to enhance their understanding of the relevant issues and encourage the adoption of sustainable construction concepts.

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Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

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